

RestaurantOS - Technical Architecture & Technology Mapping Document

Project Type

Multi-Tenant SaaS Restaurant Operating System

Primary Goal

Provide restaurants with:

- QR Based Ordering
- Kitchen Management
- Order Tracking
- Billing
- Customer Feedback
- Analytics

without requiring expensive POS hardware.

Global Technology Stack

Frontend

Technology

- Next.js 15
- TypeScript
- Tailwind CSS
- shadcn/ui

Responsibility

Build:

- Customer Dashboard
- Kitchen Dashboard
- Admin Dashboard

Reason:

- Fast UI development
 - Responsive design
 - Production-ready
-

Backend

Technology

- FastAPI
- Python 3.12+

Responsibility

Build:

- Authentication
- Orders
- Billing
- Payments
- Analytics APIs
- QR APIs

Reason:

- High performance
 - Async support
 - Easy integration with AI modules later
-

Database

Technology

- PostgreSQL

Responsibility

Store:

- Restaurants
- Users
- Tables
- Sessions
- Orders
- Payments
- Feedback

Reason:

- Enterprise-grade
 - Supports SaaS architecture
-

ORM

Technology

- SQLAlchemy
- Alembic

Responsibility

Database management and migrations.

Real-Time Communication

Technology

- WebSockets

Responsibility

Used for:

- Live Orders
 - Kitchen Updates
 - Order Tracking
 - Service Requests
-

Storage

Technology

- AWS S3

Responsibility

Store:

- Food Images
 - Restaurant Logos
 - QR Images
-

Payments

Technology

- Razorpay

Future:

- Stripe
-

Deployment

Technology

- Docker
 - Nginx
 - AWS
-

MODULE 1

Authentication & Restaurant Registration

Business Requirement

Restaurant owner should create account and login.

Features

- Registration
 - Login
 - Logout
 - Password Reset
 - JWT Authentication
-

Tech Stack

Backend:

- FastAPI
- JWT

Database:

- PostgreSQL

Frontend:

- Next.js
-

Deliverables

Restaurant can create account and access dashboard.

MODULE 2

Restaurant Profile Management

Business Requirement

Restaurant owner manages business profile.

Features

- Restaurant Name
 - Logo
 - Address
 - Phone
 - Tax Details
-

Tech Stack

Frontend:

- Next.js Forms

Backend:

- FastAPI CRUD APIs

Storage:

- AWS S3

Database:

- PostgreSQL
-

Deliverables

Restaurant profile fully manageable.

MODULE 3

Table Management

Business Requirement

Restaurant manages seating tables.

Features

- Create Table
 - Update Table
 - Delete Table
 - Capacity
 - Status
-

Tech Stack

Frontend:

- Next.js Dashboard

Backend:

- FastAPI CRUD

Database:

- PostgreSQL
-

Deliverables

Restaurant can manage tables.

MODULE 4

QR Code Generation System

Business Requirement

Every table should have unique QR code.

Features

Generate:

restaurant.com/table/{table_id}

Download QR

Print QR

Tech Stack

Backend:

- Python qrcode library

Storage:

- AWS S3

Frontend:

- Download Button
-

Deliverables

Unique QR for every table.

MODULE 5

Menu Management

Business Requirement

Restaurant should manage digital menu.

Features

- Categories
- Dish Creation
- Price Management

- Image Upload
 - Availability Toggle
-

Tech Stack

Frontend:

- Next.js

Backend:

- FastAPI CRUD

Storage:

- AWS S3

Database:

- PostgreSQL
-

Deliverables

Complete digital menu management.

MODULE 6

Table Session Management

Business Requirement

Track customer occupancy.

Features

Create Session

Close Session

Track Table Status

Tech Stack

Backend:

- FastAPI

Database:

- PostgreSQL
-

Deliverables

One active session per table.

Prevents duplicate customer sessions.

MODULE 7

Customer Ordering System

Business Requirement

Customer orders directly from QR menu.

Features

- View Menu
 - Add To Cart
 - Update Cart
 - Special Instructions
 - Place Order
-

Tech Stack

Frontend:

- Next.js Public Pages

Backend:

- FastAPI

Database:

- PostgreSQL
-

Deliverables

Customer can place order without login.

MODULE 8

Order Management System

Business Requirement

Manage full order lifecycle.

Features

Pending

Preparing

Ready

Served

Completed

Cancelled

Tech Stack

Backend:

- FastAPI

Database:

- PostgreSQL
-

Deliverables

Order workflow operational.

MODULE 9

Kitchen Dashboard

Business Requirement

Kitchen receives orders instantly.

Features

Live Orders

Order Queue

Status Updates

Special Instructions

Tech Stack

Frontend:

- Next.js

Backend:

- FastAPI

Realtime:

- WebSockets

Database:

- PostgreSQL
-

Deliverables

Kitchen receives orders in real time.

MODULE 10

Live Order Tracking

Business Requirement

Customer should track food status.

Features

Received

Preparing

Ready

Served

Tech Stack

Frontend:

- Next.js

Backend:

- FastAPI

Realtime:

- WebSockets
-

Deliverables

Customer sees live order progress.

MODULE 11

Service Request System

Business Requirement

Reduce dependency on waiters.

Features

Call Waiter

Request Water

Request Bill

Need Assistance

Tech Stack

Frontend:

- Next.js

Backend:

- FastAPI

Realtime:

- WebSockets

Database:

- PostgreSQL
-

Deliverables

Staff receives requests instantly.

MODULE 12

Billing System

Business Requirement

Generate final bill automatically.

Features

Merge all session orders

Calculate Total

Generate Bill

Close Session

Tech Stack

Backend:

- FastAPI

Database:

- PostgreSQL

Frontend:

- Next.js
-

Deliverables

Accurate bill generation.

MODULE 13

Payment System

Business Requirement

Allow online payments.

Features

UPI

Card

Net Banking

Cash Tracking

Tech Stack

Payment Gateway:

- Razorpay

Backend:

- FastAPI

Frontend:

- Next.js
-

Deliverables

Online payment workflow operational.

MODULE 14

Feedback System

Business Requirement

Collect customer satisfaction data.

Features

Food Rating

Service Rating

Ambience Rating

Reviews

Tech Stack

Frontend:

- Next.js

Backend:

- FastAPI

Database:

- PostgreSQL
-

Deliverables

Customer feedback stored and visible.

MODULE 15

Analytics Dashboard

Business Requirement

Provide business insights.

Features

Revenue Analytics

Order Analytics

Popular Dishes

Table Utilization

Customer Ratings

Tech Stack

Frontend:

- Next.js
- Recharts

Backend:

- FastAPI Aggregation APIs

Database:

- PostgreSQL
-

Deliverables

Restaurant owner can make data-driven decisions.

MODULE 16

Multi-Restaurant SaaS Architecture

Business Requirement

Support thousands of restaurants.

Features

Restaurant Isolation

Separate Menus

Separate Orders

Separate Analytics

Separate QR Codes

Tech Stack

Backend:

- FastAPI

Database:

- PostgreSQL

Authentication:

- JWT
-

Deliverables

True SaaS architecture.

MODULE 17

Production & Scaling

Business Requirement

Support large number of restaurants.

Features

Containerization

Load Balancing

Monitoring

Error Tracking

Backups

Tech Stack

Infrastructure:

- Docker
- Nginx
- AWS EC2
- AWS RDS

Monitoring:

- Sentry
- Prometheus
- Grafana

Cache:

- Redis
-

Deliverables

Production-grade scalable platform.

Future Phase (Not MVP)

AI Voice Ordering

Technology:

- Whisper

AI Waiter

Technology:

- OpenAI / Gemini APIs

Demand Forecasting

Technology:

- Prophet
- XGBoost

Revenue Forecasting

Technology:

- Machine Learning Models